

POTTERY DECORATION

Under
the
Glaze.



By
Louise McLaughlin

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Title: Pottery decoration under the glaze

Author: M. Louise McLaughlin

Release date: March 12, 2023 [eBook #70276]

Language: English

Original publication: United States: Robert Clarke & Co., 1880

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DECORATION UNDER THE GLAZE *****

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UNDER THE GLAZE.

BY

M. LOUISE McLAUGHLIN.

CINCINNATI:
ROBERT CLARKE & CO.
1880.

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PREFACE.



T has been with some hesitation that I have undertaken to write a manual on the decoration of pottery under the glaze. This feeling was caused by the fear that I should not be able to treat the subject in as comprehensive, as well as comprehensible, a manner as it deserved.

Nevertheless, in an experience of more than two years, chiefly spent in experimenting on different methods of under-glaze decoration, I have learned some things which may be of use to others, and have thought that a treatise upon the subject from an unprofessional, yet practical standpoint, might have its place in the literature of ceramics which has now become so extensive.

Let it be understood, however, that in the following pages there has been no attempt to deal with the occult mysteries of the potter's art, but simply to give an account of results attained and facts acquired, every one of which can be vouched for as a record of actual experience.

I would like here to express my sense of indebtedness to the potters who have, by their courtesy and attention, contributed to these results, and particularly to Mr. Joseph Bailey, Sr., of the pottery of Mr. Frederick Dallas, whose intelligent co-operation and valuable advice have been of the greatest service.

Walnut Hills.

Cincinnati, May, 1880.

INTRODUCTION.



THE decoration of pottery under the glaze offers opportunity for the production of work, beautiful and artistic in a high degree, and in its simpler forms gives to those who do not aspire to the higher art, a most fascinating and labor-rewarding study. I can not, however, assure those who wish to enter the domain of decorative art by this interesting road, that they will find the way strewn with roses. On the contrary, discouragements are likely to occur so frequently that it is only by the exercise of a considerable amount of patience and perseverance that any advance can be made. But such are the attractions of the work that its enthusiastic votaries will not be deterred from its prosecution by any difficulties in the way, and if immediate and perfect success can not be assured, an exceedingly interesting occupation will be revealed to the learner, and to the earnest worker can be promised the reward of faithful labor.

I would like here to insist, however, upon the necessity of a thorough and serious study of drawing as a preparation for this as well as for all decorative work, and to enter a most emphatic protest against the theory which seems to prevail in some quarters, that any one can paint upon pottery acceptably. With little preparation other than the possession of some colors, brushes, clays, and pieces of pottery, the devotee of art enters upon the work of decorating (?) pottery. Alas! disfiguring it, for it is a lamentable fact that of the pottery now being painted by amateurs (some of the work by professional decorators might also be included) a very large proportion has its value diminished rather than enhanced by the work put upon it.

This would not be so frequently the case if the scope of the work were limited to the capacity of the worker. We see every day, attempts, in which the failure to reach the point aimed at is absurdly obvious, where, if the effort had been less ambitious in its aim, the result would not only be less an object of ridicule, but of far more intrinsic value.

In other departments of human handiwork, some natural capacity, as well as a certain amount of training, is considered necessary to successful practice, but, in art, such is the ignorance of the general public, any one is thought eligible to the profession of an artist, and much passes under the name of art, which has no claim to that honorable title.

I do not wish to discourage those who are willing to enter into this pursuit as humble, patient learners, but to warn those, who are eager to obtain at once the rewards only given to years of faithful study, of the futility of their desires. It is perhaps idle, however, to warn or to advise them. The consciousness of their folly will only come to them after experience, and knowledge gained from failure, shall have given them some comprehension of the difficulties which their ignorance prevented them from seeing.

In art, as in all other pursuits, there is no royal road to excellence, and those who do not think it necessary to learn the rudiments before attempting the higher branches, will learn to their cost, if they are capable of learning any thing, that nothing was gained by leaving out these stepping stones in their progress.

The time is never misspent which is devoted to honest, well-directed efforts in acquiring the art of drawing. The excuse so often offered, that the practice of the art is only intended as an amusement, and that therefore it is unnecessary to undergo the drudgery of learning it thoroughly, is one which is unworthy. To those who offer it I would say, pray amuse yourselves with something else rather than an art, into which, to obtain even an insight, demands the best that any one can give.

These remarks apply as well to the decoration of pottery as to any other department of art work. Although from its peculiar nature and various uses this material yields itself to many and varied styles of decoration, yet this decoration, however simple, must be planned and executed according to sound principles, to be of any artistic value. The best preparation for the work of decorating pottery, therefore, is a thorough knowledge of drawing and an understanding of the laws which govern all decorative art.

Sources of information upon these subjects, drawn from competent instruction, books and periodicals, are daily becoming more numerous and accessible, and one who desires to practice the kind of decorative art we are

now considering, should cultivate his taste and augment his skill by these means, and especially by close study of nature in all its various forms. It is only in this way that the student will be enabled to accomplish work which will be a source of pleasure and profit to himself and others.

POTTERY DECORATION UNDER THE GLAZE.



CHAPTER I.

POTTERY.



E will consider in this chapter some facts in regard to pottery, which, although of a very elementary character, are not so generally understood as may be desirable. These facts relate to the substances of which pottery is composed, the processes of its manufacture, and the methods by which it is or can be decorated.

Two chemical substances, viz., silica and alumina form the basis of all pottery clays. These substances are themselves infusible (except under the compound blow-pipe), but by admixture with other materials more susceptible to the action of heat, they are made useful to the potter's art, and, when subjected to a high temperature, fuse and form the hard, insoluable combination with which we are all familiar.

Porcelain, and the finer kinds of earthen-ware, are made from clays artificially combined of various natural elements, in such a manner as to produce the qualities desired, while the coarser wares are generally made from clays in a natural or unmixed state. These clays agree in their essential characteristics, but contain other elements which cause certain differences of color, susceptibility to heat, plasticity, etc. The differences, therefore, which exist between the many kinds of pottery manufactured, may be said to be due, not so much to a diversity of the materials used, as to the changes produced by the combination of the same materials in varying proportions.

It is unnecessary that we should here enter into the details of the mixture of different pottery clays, for which each manufactory has its own formula. A few words, indicating, in a general way, the causes of certain easily-recognizable qualities, will suffice.

The peculiar beauty and translucency of fine porcelain, is due to an excess of silica, or the vitreous element of pottery, in its composition. This is attained by the use of kaolin, a fine, white clay, produced through the agency of natural causes in the decomposition of feldspar, which is itself a

silicate of alumina, and one of the principal ingredients of granitic rocks. In porcelain, it may be said, that the proportion of silica to alumina is about three to one, and the other ingredients, such as iron, potash, chalk, and soda, exist only in such quantities as are necessary to cause the fusion of the two first-mentioned elements. It is fired but slightly the first time. This leaves the body of the ware very soft and porous, and upon the application of the glaze the latter is absorbed into the body, and by its action upon the materials composing it, produces a translucent effect.

We refer, here, to French porcelain. In the manufacture of English china, the process is radically different. The last-mentioned ware is sometime spoken of as “bone china.” This term has reference to the use of bones from which the lime required in its manufacture is obtained. This element produces a translucent quality in the body of the ware independently of the action of the glaze, which in this case forms merely a coating upon the surface of the previously hard baked body.

Earthenware possesses less of the vitreous element, and the first firing is continued until the ware becomes so dense that it will not absorb the glaze, and the body remains opaque.

The term pottery, is more strictly applicable to the kind of ware last mentioned. It is to this that especial reference will be made in the following pages, in describing methods of manufacture and decoration.

Earthenware is of various colors, ranging from pure white, through the cream white and cream-colored wares to decided yellow, and lastly red. The clays can also be artificially colored, and changes in color are also produced by different degrees of temperature in firing. Ordinarily the process of firing produces a material change in the color of clay. Thus, a grey-tinted clay will assume a bright, yellow color, and a dull brown, green, or blue will be changed to a bright brick-red, under the action of the fire.

The heat used in firing should produce partial vitrification. If a piece of ware has not been fired sufficiently to produce that effect, it is said to be “soft-baked.”

White earthenware requires the greatest degree of heat in firing. In regard to the wares made from clays retaining the colors produced by the elements which they contain, in a natural state, it may be said that the amount of heat

necessary for proper firing, decreases with the depth of color. This rule, although it may accurately indicate the temperature at which any given clay should be fired, is simply artificial, except in so far as the depth of color shows the presence of an amount of foreign matter, which by its action increases the fusibility of the clay.

Red clay, being very susceptible to heat, is commonly used as a test in the firing of white ware. The same clay will change in the process from light red to dark brown, and so enable the experienced fireman to determine the degree of heat existing in the kiln.

Clays should be prepared for the making of pottery by careful sifting, dissolving in water, and, as a final precaution, straining through sieves made of fine, silk lawn. In this way all foreign particles are removed, and the clay, having attained the consistency of dough, is ready to be molded into the shapes desired. This is done in molds made of plaster of paris, or it is “thrown” by the hands of the potter on a turning-wheel, or, again, by a combination of these processes.

Before firing, the ware is said to be in the “green state,” in which condition it passes from “wet” to “hard green,” and finally to “white,” when it is ready to be fired. If fired before it has become sufficiently dry, it will be liable to crack, or scale, from the sudden expansion of the moisture confined in the clay. When dry, it is placed in “seggars,” or boxes, made of fire-clay, which are piled upon each other in a kiln, constructed for the purpose of firing the ware.

During the process of drying and firing, the clay loses a certain portion of its bulk. The amount of this shrinkage varies in different clays, but ordinarily may be said to be about one-eighth. A piece of ware, therefore, made from clay which exhibits this degree of shrinkage, would be one-eighth smaller after firing than before. The shrinkage is also modified by the degree of heat to which the clay has been subjected in firing, and there will sometimes be a perceptible difference in the size of two pieces of ware, made from the same clay, and of equal size before firing, after having been fired at different temperatures.

The length of time consumed in firing, varies with the qualities of the different wares. In the manufacture of white earthenware, the firing lasts from thirty to forty hours, while the more common kinds of ware require

less time. In kilns of ordinary size, a thousand dozen pieces of ware are frequently fired at once. Of course, considerable time is required to place the seggars containing the ware in the kiln. Some hours must elapse before the kiln and its contents will become cool enough to permit its being opened with safety, so that the firing of hard-baked pottery may require two or three days for its completion.

The first firing completed, the ware is said to be in the “biscuit,” and is then ready to be glazed. This matter of glazing is a very important one, and the success of the whole very largely depends upon the manner in which the operation is performed. To insure a successful result there must be the nicest adaptation of the materials composing the glaze to the body of the ware. The glaze for each kind of ware must be suited to its especial characteristics, and it can therefore be imagined that the number of glazes in use is very large. Each pottery has its own glaze and the variety is infinite.

The glaze used upon the finer kinds of earthenware consists of materials similar to those of which the body of the ware is made, with the addition of boracic acid (a powerful flux), which with a little lead renders the glaze fusible. In the lower grades of ware a larger amount of lead is used for a flux, and this causes the glaze to fuse at the comparatively low temperature at which these wares are fired. A glaze made of materials that fuse at an unusually low temperature is called a “soft” glaze. Some glazes are so soft that, when fired, they can be easily scratched by the point of a steel instrument. The glaze of good and durable wares is, however, so hard, that the point of the sharpest knife will make no impression on its surface.

The materials of which the glaze is made, are combined in the proper proportions and diluted to form a liquid of about the consistency and the appearance of cream. The piece of ware is then dipped carefully into the liquid, and so skillfully manipulated that it is completely covered with a coating of the glaze of the necessary thickness.

After having been covered with the glaze, which, before firing, has the appearance of an opaque white paint, the ware is ready for the second firing. This is done in what is called the “gloss” kiln, in which the heat is not brought to such a high degree as in the biscuit kiln, but is sufficient to fuse the glaze, and cause it to form a glassy, transparent surface, which

should completely cover the body of the ware, and present an equally brilliant appearance in every part.

If the glaze is not suited to the body of the ware, and does not shrink equally with it, its surface will soon present a network of fine cracks. A glaze in this condition is said to be “crazed.” This, among potters, is considered a serious fault, and in ware intended for cooking or table use, is certainly very undesirable.

Old Japanese crackle-ware, in which this condition is a distinguishing feature, is, however, much esteemed, and many, indeed, the majority of the Japanese earthenwares, both ancient and modern, display a surface of fine-crackled glaze, which may not be said to detract from their value as articles of ornament.

A similar effect is also seen in old pieces of glazed ware which have seen long service as cooking utensils, in which case it has been the natural result of the usage to which they have been subjected.

In some modern wares, both French and English, decorated under the glaze, this defect is seen. In these cases it has been caused by the use of a softer glaze than the body of the ware demanded. This expedient has been resorted to in order that the brilliancy and beauty of the colors might be preserved. These articles being intended only for ornament, beauty is more of an object than durability, and this defect of glaze may be permissible as the means of obtaining more brilliant effects. In certain methods of decoration this may be a necessity, but these cases are exceptional and experience leads me to believe that it is possible to obtain beauty of coloring with a glaze which will remain intact. It is to be hoped, however, that with the improvements in making colors the time may come when it will be possible to obtain colors which will retain their beauty under the degree of heat necessary to the production of an article which a practical potter would call a perfect piece of ware.

Pottery can be decorated either *under* the glaze, *with the glaze*, as in the case of majolica, in which the color is effected by the use of colored glazes and in the decoration of soft porcelain (*pâte-tendre*), where the painting is executed upon the unbaked glaze, and, lastly, *over* the glaze. We will concern ourselves only with the first of these methods. A few words, however, may not be out of place here, upon the differences between over

and under glaze painting, which are frequently compared and as frequently mistaken for each other.

Over-glaze painting, as is well known, is executed upon the glazed surface of the finished ware with vitrifiable colors having enough flux or fusible material in their composition to cause them to fuse at a comparatively low temperature, and so become attached to the glaze. *Under-glaze* decoration is effected by the use of colors which contain less flux, and consequently require a higher temperature for their fusion. It can be executed either upon the ware in the “green” state, or, as is more commonly the case, in the biscuit. These two methods of painting differ materially in their effects. Any one familiar with these effects can readily discover whether a piece of pottery has been decorated over or under the glaze. Those who are not practically conversant with the two methods can be easily misled, and mistakes are frequently made by those who should know better, as to the method employed in the case of a piece of ware in question.

A certain delicacy of tint and firmness of outline characterizes overglaze painting, which is easily recognized by experts, but those who can not distinguish between the two methods by this means may ascertain to which class the decoration belongs by looking aslant the surface. The glaze upon the painted portions of a piece of ware decorated over the glaze, will not, even after the most perfect firing, equal the glaze upon the uncovered portions in brilliancy, and there will probably be certain inequalities of surface between the painted and unpainted parts which will be revealed to the touch.

If the decoration has been executed under the glaze the surface will be uniformly covered with a brilliant glaze. Of the two, underglaze painting is probably the most difficult, as the colors are more liable to change under the action of the great heat to which the ware must be subjected, and the final results being, therefore, somewhat uncertain, can not be accurately counted upon, until experience has been gained from repeated failures. On the other hand, overglaze painting, while not subject to the changes produced by the fire, to so great an extent, is more difficult as regards the manipulation of the painting upon the glazed surface. The facilities for the practice of overglaze painting are greater than those afforded for painting under the glaze, which last, requires not only the handling of an artist who has

acquired facile use of the brush and some experience of pottery clays and colors, but also the assistance of an intelligent and skillful potter.

Let us not, however, laud one method of painting at the expense of the other. Both are good in their way, and confined within their proper limits, have beauties peculiarly their own. Overglaze painting possesses a delicacy of effect and a variety of color which the underglaze decoration can not rival. The latter is, however, the most artistic as well as the most effective, and lends itself more readily to the uses of decorative art and to the modern taste in color.

It is this latter method of decorating pottery which we will now consider in some of its various forms.

CHAPTER II.

COLORS FOR PAINTING UNDER THE GLAZE.



COLORS for painting under the glaze are specially prepared for the purpose from various metallic oxyds. As mentioned in the previous chapter, the colors used under the glaze have less flux or fusible matter in their composition than those used for painting overglaze: this is necessary, because in the former the colors are expected to bear a much greater degree of heat in firing than in the latter. They are so prepared as not to fuse except at the temperature required in the manufacture of the pottery to which they are applied. The variety of underglaze colors is less than that to be found in overglaze colors, as the oxyds from which colors may be made which will stand so great a degree of heat, are few in number.

The oxyds from which the colors are prepared are called the bases of these colors. The colors used in underglaze painting are made from the following bases:

Blue from the oxyd of cobalt. Co. O.

Green from the sesquioxyd of chromium. $\text{Cr}^2. \text{O}^3.$

Browns from the sesquioxyd commonly called the peroxyd of iron, $\text{Fe}^2. \text{O}^3.$, and from the sesquioxyd of manganese. $\text{Mn}^2. \text{O}^3.$

Yellow from the compound of titanium with oxygen, called titanac acid. $\text{Ti. O}^2.$

Black from the oxyd of uranium.

Red from the suboxyd, or red oxyd of copper. $\text{Cu}^2. \text{O.}$

We may add, upon the authority of M. Debette, that “pure black is obtained by taking 1 part of oxyd of uranium diluted in 22 parts of glaze; ordinary black with oxyd of manganese or oxyd of iridium; bluish black with a mixture of oxyd of cobalt and of manganese, and smoky grey with chloride of platinum. Rose is obtained by diluting, in glaze, gold which has

been dissolved in aqua regia. For the blues, oxyd of zinc and alumina may be mixed with the oxyd of cobalt. For greens may be mixed oxyd of cobalt and oxyd of chromium.”

The fluxes used for these bases are given in the following paragraphs, translated from the *Encyclopédie-Roret*:

“The matters which enter into the composition of the fluxes and which cause the adherence of the metallic oxyds are quartz, feldspar, borax and boracic acid, nitre, the carbonates of potash and of soda, red lead and litharge, and oxyd of bismuth. At Sevres, they employ, for under-glaze painting, seven kinds of fluxes, which suffice for all the colors. The majority of these fluxes are composed of quartz, oxyd of red lead and of boracic acid, and to some is added a small quantity of carbonate of soda.

“The seven fluxes of Sevres are, first, the flint flux, which is prepared by melting quickly in a crucible and then flowing upon a metallic plate a mixture of 3 parts of red lead or of litharge, and 1 part of Etampes sand.

“Second. The flux of greys prepared with 6 parts of red lead or litharge, 2 parts of Etampes sand, and 1 part of pulverized borax.

“Third. The flux of carmines is made of 1 part red lead, 3 parts of Etampes sand, and 5 parts of pulverized borax.

“Fourth. The flux of purple is made with 3 parts of red lead or of litharge, 1 part of Etampes sand, and 5 parts of crystallized boracic acid.

“Fifth. The flux of violets is made of 27 parts of litharge or red lead, 2 parts of Etampes sand, and 11 parts of crystallized boracic acid.

“Sixth. The flux of greens employed, as well as the two preceding M. Salvetat, was prepared by him with 8 1-9 parts of red lead or litharge, 1 part of Etampes sand, and 2 parts of crystallized boracic acid.

“Seventh. Last the flux of the metallic substances, which is of sub-nitrate of bismuth, obtained by decomposing in water the nitrate of the acid of bismuth, to which is added 1-12 of borax.”

These fluxes are mixed with the basic oxyds in greater or less proportion, according as the colors are intended for work, which will require a greater or less degree of heat in firing.

Underglaze colors in powder suitable for painting on pottery, can be procured of several different manufactures. As mentioned before, the variety of these colors is not so great as is to be found in overglaze colors, but as they can be readily mixed the number is sufficient for all practical purposes.

In my own work I have made almost exclusive use of French colors, those manufactured by M. Lacroix. These colors are finely ground and of considerable variety of tint. A list of those which have been tried and found to give satisfactory results, may be given here:

Blues—*Bleu de Roi* (King's blue).
Bleu violacé (violet blue).

Light blue—*Bleu myosotis* (Forget-me-not blue).

Reds—*Rouge T.* (Red T.) and *Rouge P. W.*

Carmines—*Rose* (or Pink).

Greens—*Vert foncé*, No. 1 (dark green).
Vert tendre (light green).

Black—*Noir*.

Browns—*Brun foncé* (dark brown).
Brun No. 5 (Brown No. 5).
Brun jaune (yellow brown).

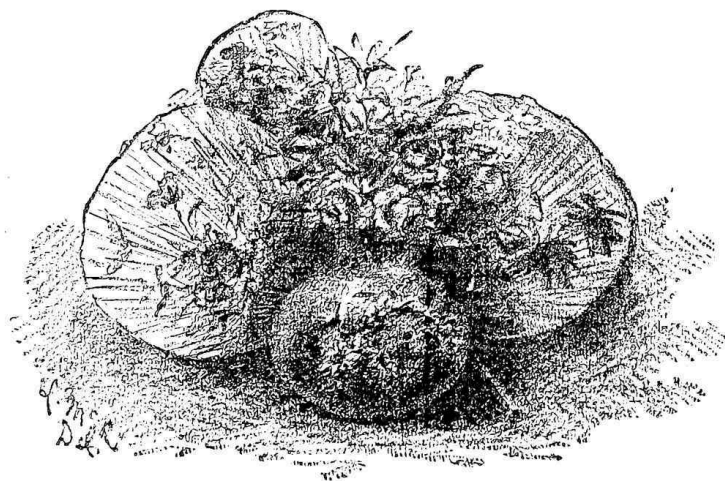
Yellow—*Jaune foncé* (dark yellow).

Grey—*Gris clair* (light grey).

There are many others, but these are mentioned as among those that have been tried and found to fire well, and the colors given in this list will be sufficient for the production of all the tints needed in underglaze painting. In faience painted after the Haviland method, fine white clay takes the place of white.

In painting on the biscuit white paint is sometimes used. This can be procured of English manufacture, but is not, I believe, made by M. Lacroix. The colors manufactured by Messrs. Hancock & Sons, Worcester, England, are also very satisfactory. The tints of these and the colors of other English manufacturers corresponds with those of M. Lacroix, which have been mentioned. The latter manufacturer, however, furnishes a much greater variety of colors, and the reds especially are superior to any English reds I have seen. M. Lacroix's list includes at least three reds, one of which, *Rouge T.*, can be relied upon to produce as good a scarlet as is possible under the glaze, when used under the proper conditions. The English carmines are very satisfactory. The English green, called French green, corresponds in tint to that of M. Lacroix, called *Vert foncé*, No. 1, while mazarine blue, of the Worcester and Phillips manufactures, and Cobalt blue of others corresponds to the *Bleu de Roi* of the French, and for work on biscuit white ware stands the fire rather better. The Victoria green, of the Worcester colors, is especially to be commended for a light green which stands the fire remarkably well.

Phillips' English colors are also very good. His mazarine blue, especially, is one of the finest blues I have ever met with. Emery's and Harrison's colors fire very well, but are not so finely ground as the first mentioned.



CHAPTER III.

PAINTING ON POTTERY AFTER THE MANNER OF THE HAVILAND OR LIMOGES FAIENCE.



HIS method of painting on pottery is said to have been discovered by M. Laurin, at *Bourg-la-reine*, in the year 1873. The process was afterward adopted by M. Haviland, and by him improved in such manner as to materially change the method, as well as the appearance of the painting. Specimens of the ware were first exhibited at the Centennial Exhibition, in Philadelphia, in 1876. The first application of a similar style in the decoration of pottery, in this country, was made by the writer in Cincinnati, in October, 1877.

This method of decorating pottery, although not involving the use of any new principle, was yet so entirely novel an application of principles already in use, as to entitle M. Laurin to all the credit attaching to a very original and important discovery. It places in the hands of the painter of pottery a method at once so artistic, and so thoroughly in accord with the modern school, as to awaken a profound interest in the minds of all lovers of art. It is probable that there are capabilities in this art that have not yet been brought out. It is still in its infancy, and that there are in it possibilities of much importance to the artistic world, can hardly be doubted.

If it were not for the technical difficulties which surround all work on pottery, and this style in particular, it would offer facilities for the production of works of art unequaled by any method heretofore in use. It is the hope of the writer that the following description of the method of decorating pottery, in this manner, may throw some light upon these technical difficulties, and also that artists of ability may be induced to try it, and so demonstrate the capability it undoubtedly possesses.

The mere knowledge of the materials used will, however, no more produce artistic work, than a box of Winsor & Newton's colors, in the hands of a beginner, will enable him to paint a picture, equal to one by Titian. To

produce good work in this method, there must be a certain amount of skill at the command of the painter, just as the same degree of skill is requisite in the production of a good picture by any other method. There is a certain boldness of effect produced by the very nature of the materials and process, which probably would not be seen in the work of the same person in other methods, yet the lack of artistic feeling and ability will be as painfully apparent in this as in any other.

Colors may be daubed upon pottery, as they are, alas! upon canvas, by those whose training and whose feeling for art would hardly fit them to become good house painters; but the result will not be good art, nor will it ever be its own excuse for being. If other branches of decorative art require taste, knowledge, and practical skill, so much the more does this, when it offers scope for the highest capacity. To the artist of ability sufficient to make use of it, it furnishes a palette which, although not of the same range as that of oil colors, yet affords an almost unlimited scale of colors, each of which is enhanced to the fullest degree by the brilliant glaze, with which the work is finished. The painting executed with these beautiful colors, moreover, is practically unchangeable, and none of the ravages of time, short of the destruction of the piece of ware itself, can affect it. In decorations for buildings, or for ordinary use in portraiture, or the higher forms of art, it offers, what has long been desired among artists and art lovers, a method of making works of art indestructible and beyond the possibility of change.

The ware known among dealers under the name of “barbotine,” has some resemblance to that which we have been considering. It has a light body, which has been subjected to a very slight fire, and is covered with a soft glaze, which ensures great brilliancy of coloring. The ware, however, has no durability, and is a substance that would not stand the action of the elements, if used in external decoration.



CHAPTER IV.

METHOD OF PAINTING FAIENCE UNDER THE GLAZE WITH THE USE OF BODY COLOR.



HE method of decorating pottery here described is similar in its effects to what is known as the Haviland, or Limoges faience, and is given as the result of numerous experiments made by the writer.

The finished work presents the appearance of a painting in oil, to which a brilliant glaze has been applied. This glaze not only renders the colors unchangeable, but gives a beauty and effectiveness which could be acquired in no other way. There is nothing peculiar in regard to this glaze, however, the only requisite being that it should be suited to the body of the ware, and that the latter and its glaze must not require so great a degree of heat in firing that the colors shall be injured. This matter of the glaze has been almost uniformly misrepresented in accounts of the ware which have been published. It has been said that the peculiar effect of the work was due to the glaze, and that the secret of making it was not generally known. This is not at all true, as the results are due solely to the peculiar method of painting, and the glaze is simply the process by which it is finished, and bears the same relation to it as the varnish does to the painting in oil. The work will suffer, of course, if this part of it is not well performed; but the distinguishing feature of the method consists not in the glaze, but in the use of clay, which is mixed with coloring oxides, capable of bearing a high degree of heat in firing, and which gives them body, producing a thick *impasto* in the painting. The work partakes, therefore, both of the nature of painting and modeling, as the high lights may be laid in so thickly as to produce an actual relief.

It is somewhat difficult, indeed, almost impossible, to give a correct idea of the palettes to be used in this kind of painting. If colors could be procured, already prepared for use, as oil paints are, and these colors had the same appearance after firing as before, it would be comparatively an

easy task. As it is, the colors must be mixed with clays in certain proportions, and, on account of the change produced by firing, the proportions necessary to produce the intensity of color desired, can only be determined from experience. The result, after the work is finished, differs from its appearance before firing to a greater extent than in any other kind of decoration upon pottery. As a rule, it may be said that the colors are intensified by firing. This is also true of other kinds of under-glaze painting, but not in so great a degree as in the case in question. The harmonies and contrasts of color can be kept only in the mind of the artist, and every part of the work must be done with a view to the result when finished, which, as has been said, will differ materially from its appearance during the progress of the painting. Experience only, can give an accurate idea of these changes. This, however, is not an insurmountable difficulty; care and patience added to the requisite artistic ability, will soon lead to satisfactory results.

Before describing the method of painting, it may be well to consider the kind of clay of which the body of the ware should be made, and the state in which the painting should be applied. As to the clay forming the body of the ware, it must be of such a nature as to adapt itself to a glaze sufficiently soft to preserve the colors. As the clay used in the painting must, of necessity, in most cases, be white, in order that the purity and beauty of the colors may not be affected by admixture with it, a body of the same, or very nearly resembling it, would, in some respects, be the best, one reason for this being that the applied clay would be more certain to adhere firmly to the body, both having the same qualities. The proper glaze for such a body would, however, require too great a degree of heat in the firing, and none but the strongest colors could bear it without injury.

In order, therefore, that the glaze may be perfectly adapted to the body of the ware, and yet require no greater degree of heat than the colors used in the painting will bear, it is best to use a body formed from materials which do not need to be fired at a very high temperature. This desired quality is found in some of the natural colored clays. A mixture partaking of the qualities of both yellow and red clay, without the unpleasant color of the former, and possessing greater strength than the latter, has been found to be the best.

THE STATE OF THE WARE IN WHICH THE PAINTING SHOULD BE APPLIED.

For this method of decoration the ware should be in the “green” state. The more moisture it contains the better, and it will be in proper condition if procured as soon as it leaves the hands of the molder, or as soon as it has become sufficiently dry to retain its shape. Pieces can be kept in this state a considerable length of time, by placing them in a box which has been lined with plaster of paris. This lining should be about an inch thick, and if the inside is occasionally sprinkled with water, it will remain moist, and keep the clay in good condition. A box of this kind is really invaluable for all work, where it is necessary to keep clay moist. Within certain limits the clay is improved rather than injured by keeping in this way; the effect upon it being that it becomes tougher and more plastic. If kept too long, it becomes “rotten,” and the plastic quality is lost, but it will keep a reasonable length of time, even two or three months. If the box can not be had, the pieces may be covered by a rubber cloth, which should be kept from touching them by a frame of wood or galvanized iron wire, damp cloths having been placed around them to produce sufficient moisture.

It is necessary to exercise this care in keeping the ware moist, in order to prevent the clay, which is applied to the surface in painting, from cracking during the process of drying. It must always be borne in mind that the material made use of in painting is simply clay, which has been artificially colored, and, as clay, becomes subject to the rules which govern all work of this kind, whether it is making pottery, modeling, or painting faience.

To produce depth and richness of effect, the painting should have a heavy *impasto*, and the high lights should be laid on in such a manner as almost to present the appearance of modeling in relief. When this is done, there is a tendency in the applied clay to crack in drying, if the conditions are not right. In applying one piece of clay to another, it is requisite that both should be of the same consistency, to ensure equal contraction. In this case, however, the clay is used as paint, and must be of such consistency as will permit its manipulation with the brush. Its application, then, to a body of clay that has lost much of its moisture, is attended with some risk, and the fact that the coating of clay is comparatively thin, alone renders it possible to defy one of the conditions which govern the use of this material. When

the piece of ware, upon which this painting is applied, has previously become too dry, the moisture in the coating of clay is very quickly absorbed into the body, and this causes it to shrink unequally with the body, and thus to crack. When the body and the applied clay are more nearly in the same condition, and the former still retains considerable moisture, it does not absorb that of the clay on the surface so rapidly, and they shrink and dry together.

The drying of the clay used in painting can, however, be retarded by the use of gum tragacanth, which may be mixed with it. It would, perhaps, be better always to take the precaution to use this medium, although when, as remarked before, the body of the ware is still quite soft, and the condition of the atmosphere such as to prevent too rapid drying, the painting may be executed simply with the use of water.

Just here a caution as to the use of gum tragacanth may be given, and that is, that it is better not to mix the gum with more of the colored clay prepared for painting than is to be used at once. When mixed only with water, the clay after becoming dry, can be easily rubbed down again with water; but if gum tragacanth has been used in the mixture, and it has been allowed to dry upon the palette, the gum will not re-dissolve readily under the action of water and the palette knife, and the mixture will be full of small particles of undissolved gum, which will render it unpleasant to work with. To prepare gum tragacanth for use, it is dissolved in water, to which may be added a drop of salicylic acid to keep it from turning sour.

The clay, which is to form a body for the colors, must be dissolved in water until of a proper consistency for painting. It should be a little thicker than the diluted clay used by potters and technically called "slip." As to the quality of the clay, the desideratum is that it shall be colorless and not inclined to crack in drying. The clay used in the manufacture of white granite ware is probably the best clay for this purpose, generally accessible. This can be mixed with all the colors without injury to the tint of any. Parian clay produces the most beautiful effects, and, where it is possible to procure it, forms the best medium for this painting. More care must, however, be exercised in its use, as it is very liable to crack in drying, and from the translucency produced in the body by the action of the fire, must be laid on more thickly than clays which have greater opacity, to produce the effect desired.

In the case of yellow, which does not always stand the fire well, it is best to mix the color with a yellow-tinted clay to ensure a successful result. C. C. clay, which is a cream-tinted white, will effect this. If a darker yellow is desired, clay which in firing acquires a decided yellow-tint may be used.

A light cream-tinted clay, such as C. C., or “stilt” clay, may be used with yellows therefore, with advantage, with browns and blues without disagreeable effects; but in the case of scarlet, pink, or crimson and greens, nothing but white clay must be used, if purity of tint is desired. The reason for this is that the presence of iron in the yellow-tinted clays injures the colors mentioned.

The clay can be procured from the pottery in broken pieces, or, better, in a semi-pulverized state from the trimming in making the wares. When in this state it is easy at any time to dissolve enough for use. It is, perhaps, better to prepare it just before it is to be used, as the longer clay stands, after it has been mixed with water, the more compact it becomes. When it has just been dissolved, it still retains air among its particles, and this renders the mixture more porous and less liable to crack. It soon dissolves in water and can be stirred until smooth. It may be strained through a sponge, alternately allowing the sponge to absorb the diluted clay and then squeezing it. Probably the most convenient way is to rub the clay in a mortar until all the lumps are dissolved. If a very small quantity only is needed it may be rubbed down upon the palette with the palette knife as it is wanted.

A palette or a slab of glass may be used for mixing the colors and clay. The latter is the best, as it can be procured of large size, and as in this kind of work a considerable quantity of paint is necessary, there should be ample room to mix the colors. A slab of French plate glass, at least twenty by twenty-four inches, may be procured. This can be imbedded in plaster of paris, which will make it stronger, as well as furnish a white surface upon which the colors will show to better advantage.

The powdered colors as procured are not perfectly pulverized, and must be well rubbed down with a muller and palette knife. It is even more necessary in this than in other kinds of painting upon pottery that the color should be well ground. If not rubbed down sufficiently *before being mixed with the clay*, particles of color which are scarcely apparent before firing,

will show when the work is glazed, forming specks in what should be, a perfectly uniform tint. If the color is mixed with the clay before it has been perfectly dissolved, it will not afterward be possible by any rubbing down to remove the specks from the mixture.

The colors, after being thoroughly ground, can be mixed with clay in such proportions as may be desired. The white clay takes the place of white, and is used to lighten every tint, as well as to form a body for the colors. It is a good plan, where experience has rendered it possible to do it with judgment and certainty of result, to mix the various tints in water with clay, in quantities, and then allow them to dry. When dry, they can be pulverized and the colored powder can be put away in bottles for future use. This saves time when painting, and also ensures a like result from the use of a mixture which has already been tested and found to be satisfactory. It is only, however, after considerable experience in mixing colors, and verifying their results, that this can be safely done.

As in other kinds of underglaze painting, it may be given as a general rule, that the colors are intensified in firing, in this, however, much more than in other methods. All the stronger colors are intensified to a degree difficult to realize before experience has taught the artist what to expect. They all become several shades darker, and, in the case of blues, blacks and greens, it is more difficult to obtain delicacy, than depth of tint.

In the use of scarlet, crimson and yellow, a considerable amount of the color in proportion to the clay must be used for the darker portions to give sufficient depth. Only enough clay must be taken to give body to the paint, and the mixture should then be applied very thickly. The painter must also bear in mind the fact that the stronger color will overpower the weaker in mixtures, even when the latter is in such degree as to give the color to the mixture before firing. After firing, the stronger color will appear in excess. In making such mixtures, therefore, the stronger color should be used very sparingly, if it is desired that the other should predominate in the tint when fired. Otherwise, the colors may be found after firing to be entirely different from what was intended.

METHOD OF PAINTING.

Let us suppose that the artist wishes to paint a vase. A certain tint being selected for the ground, the color, or colors, which are to produce it are mixed in their relative proportions, according to the hints given above. A sufficient quantity should be prepared to paint the whole ground, especially if a mixed tint, which it would be difficult to reproduce in its exact proportions. It will require some little experience to enable the artist to judge how much will be needed, and, perhaps, it will be somewhat difficult for any one accustomed to painting in other methods, to realize the quantity of paint used in this. It is better to have too much mixed than too little, as it can be kept and used another time.

The color, or colors, for the ground having been rubbed down until perfectly smooth, as directed before, may be mixed with more or less clay, according to the shade of color desired. The first tint may be made to represent the darkest shade in the ground, and a comparatively small amount of clay should be used. After the color has been mixed with the clay, it can be placed on one side of the palette. By taking portions from it and adding more clay, other shades of the same color may be made. First, a light tint may be prepared for the first coat upon the vase.

Before commencing to lay on the ground, it is well to wash the surface of the vase with a thin solution of glaze (such as is used for finishing the ware), or with borax water. This is to ensure the adhesion of the clay, of which the painting is to consist, to the clay of which the vase is made. Previous to this, the vase, if not already in good condition, should be washed over with a brush, or with a sponge dipped in water, or sprinkled until it has absorbed water enough to keep the clay, which is to be applied to its surface, from drying too quickly.

A broad, flat camel's-hair brush may then be charged with the *light tint*, and the surface of the vase covered with it as evenly as possible, and so thickly as to completely obscure the body. This done, another tint, darker than the first, in a degree sufficient to permit their being distinguished from each other, may be mixed. This will form the middle tint of the ground, and is to be laid on over the first. The reason for applying two coats is, that, although it might be possible to paint one with the degree of thickness necessary to prevent the shrinkage of the applied clay in the firing, from revealing the body of the vase, still the beginner is very likely to be deceived as to the thickness of the *impasto*, judging by the eye alone.

Potters dip pieces of ware in colored slips, and the thickness of the covering thus formed is all sufficient. In the case of painting, however, it is impossible to lay the clay on as evenly, and there will be, in all probability, some places too thinly covered to stand the fire, but which would not be revealed before. It is better, therefore, to apply two coats, so that one may cover up the deficiencies of the other. The reason for having these two coats of different tints is, that it is then possible to be assured that the surface is covered completely, it being easy to distinguish the first tint laid from the body of the vase, and in the second painting, to observe that it entirely covers the first.

The second coat finished, a lighter tint should be mixed, with which the ground is varied by touches here and there, making the highest light of the ground, while touches of the first tint mixed, give the darker shades. There should be difference enough in tone between the middle tints of the ground, and these lighter and darker tints, to produce a good effect of light and shade, and these touches should be laid on with a free hand, and then softened into the ground, care being taken that the lights and shadows should not end too abruptly.

In these touches, as in all the painting, there must be a certain thickness in reality, as well as in appearance, to make the work effective when finished. The clay shrinks and seems to be dissipated in the process of firing, and if applied too thinly, in any part, will reveal the body of the ware, or some under-tint, perhaps spoiling the appearance of the whole. The transparent effect, produced by a thin layer of clay, which, after firing, shows the under-tint through, is sometimes very beautiful in certain parts of a design. It is, however, difficult to judge by the eye of the requisite degree of thickness necessary to produce this effect, and the heat of the firing may be so great as to cause that portion of the painting to disappear entirely. It can not, then, be safely counted upon, and the best way is to depend solely upon the lights and shades of the painting for the effects desired.

The ground finished, the decorative design may be painted upon it. This should not be outlined upon the surface, but should simply be painted with a free hand, and without too much attention to detail, a brilliant effect of light, shade, and color being the object aimed at in this style of painting.

We will suppose that a floral design is to be painted upon the vase. The middle tint of the flowers can first be laid, the shadows are then put in, and lastly the high lights, laid on heavily, almost giving the effect of relief. Leaves and other accessories of the design may be treated in the same manner. The edge of the design must be softened into the background, to avoid a hard effect after the work is glazed. The process of glazing has a tendency to soften and melt the tints into each other, but this effect must be enhanced by judicious use of the middle tints and shadows of the painting, and by leaving the edges thin. One of the greatest beauties of this kind of painting, when well done, is the effect of the rich colors melting into each other, with a charming indistinctness, which leaves something to the imagination of the beholder.

Those who have not been accustomed to the use of color in such masses, will, perhaps, be embarrassed at first by the difficulty of painting with clay. This will soon be overcome by practice, but, as has been said before, it should always be remembered that it is clay which is used, and, therefore, the work is subject to the conditions which govern the use of that material.

It is unnecessary to enter further into details, as it is only intended here, to give some instruction as to the method of painting, leaving the manner of it to the artist, it being taken for granted that any one desiring to practice this kind of painting should have already attained some proficiency in the use of colors, and should have acquired a knowledge of the rules which govern art.

To such there will simply be the difficulty of learning to work with a new material. If this knowledge does not exist, it will be no easier to succeed in accomplishing any thing, worth the doing, in this, than in other kinds of art work.

With regard to the colors to be used, it is enough to say that a sufficient number can be obtained to produce by admixture, in various degrees, tints for the production of any subject required. It is only necessary to remember, as said before, that in these mixtures the stronger colors must not be in such proportion as to overpower the weaker. A list of useful colors has been given in a previous chapter. In regard to the mixtures of these colors a few words may be of use.

Black is a very useful color for shadows, making, in its admixture with white clay, numerous pleasing tints of grey, which may be varied by the addition of any other color which may be desired. As black is one of the strongest colors, however, it must be very sparingly used where a light tint is needed. In the mixture of blue and green, the former must exist in a much smaller proportion than the latter, if the mixture is to have a greenish tint. Experience alone can give success, but enough has been said to prevent the beginner from making very disastrous mistakes.

The painting must, of course, be executed while the clay is wet, and should be finished in one sitting. If this is not possible, it may be kept wet in the way described for keeping the ware before painting.

In describing the method of painting a vase, we have taken for our model, one decorated with flowers. Decorations can, however, be as varied in regard to subjects as paintings on canvas. Floral decorations seem to be the best adapted to vases, or other objects having rounded surfaces. Plaques and vases, such as pilgrim jars, which afford flat surfaces, may also be decorated with landscapes or figures. For these subjects somewhat different treatment may be adopted, but enough has been said to indicate the requirements of the material, and each artist will find a manner suited to himself.

Decorations of a similar character can be produced upon ware of different colors, leaving the color of the clay for a ground. This has a very good effect when done upon cream-tinted clay in suitable colors. As this kind of ware should be fired at a higher temperature than the deeper tinted and coarser clays, the colors are liable to fade in the firing. This is not undesirable, however, as the faded tints harmonize with the light ground much better than darker colors. Some beautiful effects can be produced by decoration of this kind, which can be supplemented by carving and gilding.

The same method can be used on a ground of dark red clay by the selection of colors which will harmonize with the ground. Yellow flowers look well upon a ground of this character, and the yellow color can be produced, as we have said before, by the use of yellow clay, either alone or as an adjunct to the color.

Monochrome decorations can also be produced, by the use of natural clays of various colors, in the same manner as that pursued in the use of the

artificially tinted clay of which we have been treating. Good effects can be produced in this way, with the advantage that the colors will be fully retained in the firing. Red, or brown and white, in various degrees of combination, can be used to produce a design with every effect of light and shade. Also, red, with cream-tinted clay, or the three colors—red, yellow and white—can be used together. These designs can be produced upon grounds of cream, red, or yellow clay.

DRYING.

When the painting has been completed, it should dry very slowly, and it would be better, if possible, to effect this by placing it in a moist, cool place. It is, however, difficult to handle a piece of ware when wet. The clay is in a very soft state, and a sudden jar might cause the vase to fall to pieces.

If the vase has not been in proper condition for painting, or has dried too quickly, it will, after some hours have elapsed, begin to show fine cracks upon the surface of the painting. These may be stopped, before they have gone too far, by passing a modeling tool over them, or, if the crack has become too deep for this treatment, it may be filled with clay as nearly as possible in the state at which the body of the ware has arrived. If the cracks are allowed to go too far it will be difficult to stop them, as they may have extended into the body of the ware.

If the piece is permitted to go to the firing with any cracks, however small upon its surface, they will become widened in the firing, and, especially after the glazing, will show very distinctly.

In the biscuit, they may be stopped by filling them with powdered clay mixed with gum water. But an ounce of prevention is worth a pound of cure, and the safest method is to prevent their appearance in the first place. If this can not be done, they must be stopped as soon as they appear. The finer clays seem to have a greater tendency to crack in drying, and when the cracks have appeared, they are more difficult to stop than in other clays.

FIRING.

When the piece decorated is thoroughly dry, it should be fired at a temperature sufficient to make the body durable, and at the same time to perfectly fix the colors of the painting, so that there may be no danger that the glaze will cause them to run in the final firing. This latter defect may also result from an excess of glaze, and where it exists only in a very slight degree, produces a melting softness which is very pleasing. This effect is, however, so likely to be exaggerated to the ruin of the work that it is best not to run any risk of the kind in firing, and to depend upon the painting alone for the harmonious blending of the tints.

The temperature at which the work is to be fired, must, of course, depend upon the qualities of the clay of which the piece of ware is made. As has been said before, this clay should be one that does not require an excessive degree of heat in firing. The exact degree can be estimated by the potter who understands the requirements of the clay which has been used in the manufacture of the particular piece of ware in question. It may be said that the temperature at which Rockingham ware is fired is suitable for this work, and that a glaze such as that used upon that kind of ware can be used with good results. It is possible to have a glaze so adapted to the body of the ware as to prevent it from crazing, and yet preserve the brilliancy of the colors.

CHAPTER V.

MODELING IN RELIEF.



UNDER this head may be comprised the decorative modeling of figures and other designs upon pottery, in various degrees of relief. First, we have figures in relief forming decorations upon vases. These, as well as other designs in high relief, belong properly to the domain of sculpture, and the methods of producing them do not vary essentially from those in use in that art. The final process in this case is the firing, instead of the casting in plaster and cutting in marble.

Of the methods of producing floral designs in relief, a few suggestions may be of use. These, as well as the painting described in the previous chapter, must be executed when the body of clay, to which they are applied, is very wet, and the applied clay must in this case be, as nearly as possible, in the same condition. If the clay is not in the proper condition, the work will crack in drying.

The vase which is to be decorated being in the right state, a lump of clay, having the plasticity necessary for modeling, is procured. This may be of C. C., a deeper cream color, or red, according to the color desired. White granite is not plastic enough, and is, besides, liable to crack in drying. But of all white clays, parian forms the most beautiful substance for this kind of modeling. Leaves and petals of flowers can be shaped from pieces of clay, which have been flattened to the required thickness, either by pressing upon a slab of plaster of paris, by means of a weight, such as potters use; or, if the work is not on a large scale, the clay can be shaped in the palm of the hand. The forms can then be cut with a sharp modeling tool. If there is more moisture in these pieces of clay than in that which forms the body of the vase, they should be allowed to remain a short time upon a dry plaster slab. The plaster will quickly absorb the moisture in the clay. When ready, the pieces can be placed in position on the vase, the places where they are to be applied having been previously washed with clay and water (technically called “slip”), or with glaze water, to make them adhere. Stems are made by

rolling pieces of clay between the palms of the hands, or upon a flat surface of wet plaster or wood. They are afterwards allowed to dry, and when in proper condition, applied in the same manner as the flowers and leaves. Delicate touches may be put in with a brush dipped in clay, of a proper consistency for painting. The forms of the design can be shaped and finished after they have been applied to the vase by the use of modeling tools, and, finally, with a camel's-hair brush. The design may also be brushed over with a sponge, moistened with water, which will also be found very useful in restoring the smooth surface of the ground, if it has been injured in the process of modeling.

After the modeling has been completed, the work should be gone over with a modeling tool, and in places where a mass of clay has been attached to the surface, the edges should be under-cut. This will prevent cracking. If this precaution is not taken, a crack thus produced may extend into the body of the vase and render it unfit for firing. Where a large mass of clay occurs in the modeled design, it is better to hollow out the projection from the inside of the vase, to prevent the cracking or warping of the whole in firing, caused by the unequal thickness.

Cracks, which may appear in drying, should be filled, or, if possible, be pressed out with a modeling tool, before the piece is fired. If the crack is so deep that it is necessary to fill it, the place should be cut open with a modeling tool, moistened with water, and filled with clay, as nearly as possible in the same condition as that upon the vase.

Flowers can be modeled in parian clay upon a creamy-tinted ground with good effect. Various colors of deeper tinted natural clays, or clay which has been artificially colored in the body, can be used as the ground of such decorations. Light grounds are sometimes colored after the first firing, in the biscuit; or a ground can be painted in the method described in the previous chapter, while the vase is soft, and the modeled decoration can then be applied instead of the painting.

Decorations of this character in high relief are frequently colored in natural tints, either under the glaze, in the biscuit; or over the glaze. The taste of this manner of finishing relief-work is somewhat questionable, and the same objection might apply to it as to colored sculpture. That it has received the sanction of the best factories in the world is no proof of its

being in accord with correct taste, as the many (artistic) atrocities of Dresden and Capo di Monte testify. Some of the later Haviland faience, however, having a ground in the usual style of that ware with decoration in high relief, colored with subdued tints, harmonizing with the grounds, is so artistically done that one can hardly think it open to this objection, although it can not be considered an improvement upon the earlier work of this establishment in which the colored decorations were simply painted upon the surface.

It is only to the use of colors which imitate those of nature upon modeled decorations that there could be any objection. Relief-work may be conventional in its form, or colored conventionally, in one color or many, or gilded, with perfect taste. The latter method produces a fine effect, but, to make it successful, the color of the ground should be well chosen, and the design simple in form and not too much detached from the surface. The gilding looks best when left unburnished, and should only be rubbed enough to give the effect of dead gold.

Next, we have applied designs in low relief. These are made in several different ways, according to the effects desired. Work of this kind is more commonly executed by the method used in the production of Wedgwood-ware. The designs to be applied are first modeled in clay. Casts are then taken of the models in plaster of paris. Clay of the kind desired for the relief designs is then pressed in the mold and the edges trimmed off. The design can be removed from the mold by the aid of a flat, steel instrument, with a short, broad blade; or a palette knife, which when pressed upon the clay adheres to it and detaches it from the mold. It can then be transferred to the piece of ware to be decorated, to which it is made to adhere by the use of "slip," as described before.

This method is very useful, indeed, indispensable in the manufacture of such wares commercially, when the same designs are to be repeated many times, but it is hardly to be recommended for artistic work, except for conventional designs, borders, etc. This method of casting the forms in a mold may be useful, however, in the reproduction of leaves of intricate outline, which can be made of various sizes, and, from the plasticity of the clay, can be infinitely varied in position according to the taste of the worker.

When objects of delicate and intricate form, such as threads, spider's webs, nets, and sometimes leaves, are to be represented in work of this character, it can be accomplished by dipping threads, or nets of cotton, or, if leaves, the natural leaves themselves, into "slip." The clay adheres to the threads, or forms, and after it has set sufficiently, they can be placed upon the object to be decorated. On firing, the substance inclosed within the clay is dissipated and the clay only, remains. Some very wonderful and beautiful effects may be produced in this way; but, as these delicate forms are necessarily very fragile, especially if much detached from the surface, they are only suitable for articles intended solely for ornament.

Designs in low relief can also be applied with the brush in the forms desired, which may afterward be shaped and finished by the use of modeling tools, or, when dry, carved with steel instruments. This latter is said to be the method by which M. Solon executes his exquisite designs in fine white clay upon ware which has been colored in the body. One of the greatest beauties of M. Solon's work is the transparent effect produced where the clay has been left so thin as to show the colored ground through.

Work in the very slightest relief can be done with the brush alone. Care must be observed in this, however, that the application of clay is not so very slight as to produce complete transparency upon firing. By a judicious mixture of work in which the clay is so heavily laid on as to remain opaque after the firing, and of that which is applied so thinly as to become transparent, very delicate and beautiful effects can be produced by the use of white clay upon colored grounds.

The colored grounds in the Wedgwood and Solon ware are produced by the use of a fine clay body, which has been artificially colored throughout, which, in the case of Wedgwood, is finished by a very slight glaze, and in the Solon ware, by a full glaze, by which the transparent effects of certain parts of the work are brought out. The colors used in the body of these wares are blues, dark greens, and olives, and also black. Where it is not possible for the amateur, for whom these hints are written, to obtain such wares for decoration, very pretty work can be produced, as has been mentioned before, by the use of a body of naturally colored clay upon which white or a light-colored clay can be applied.

Relief-work can be finished with a full glaze, a very slight glaze, or can be left unglazed. As a general thing it may be said that the modeling in a design in high relief will look best with but a slight glaze, or none at all. The slight glaze is called by potters a “smear” glaze, and is effected by firing the work in a “seggar,” which is glazed upon the inside. The unglazed ware absorbs enough glaze from the glazed surface to give it a semi-glazed appearance, which is very pleasing. Work in partial relief, as well as articles in full relief, can also be finished by the application of colored glazes. This is the method by which the ware known at the present time as majolica is produced. Blue and green glazes of this kind are frequently to be obtained in potteries in this country, where they may be applied to work which has been modeled. Other colors in majolica glazes are to be had from the dealers, and if the colors desired can not be applied at the pottery, the artist can apply them with a brush in the same manner as paint is used upon the biscuit. The effect of work of this kind can also be enhanced by gilding put on over the glaze.

CHAPTER VI.

INCISING AND CARVING IN CLAY.



THESE are two of the simplest as well as most effective methods of decorating pottery. We will treat first of incised work in clay.

The piece of ware to be decorated must, of course, be in the “green” state; not too wet, yet not dry enough to be brittle. An outline drawing can then be executed upon its surface with a pointed modeling tool, not too sharp, and perfectly rounded at the point. If any burr is thrown up in the course of the lines, it must not be brushed off until the piece has become partially dry. The drawing can then be brushed over with a stiff camel’s-hair brush, leaving the lines free. These lines can now be filled with dark (under-glaze) paint (black is best), mixed with a little gum arabic and water. In going over the lines some of the paint will extend over the edges; this can be scraped off, when the clay is dry, with the edge of a sharp steel blade, two or three inches long, held between the fingers and thumb. A broken knife-blade will do for the purpose. This scraping must be very lightly and carefully done, to avoid injuring the surface of the ware. The surface may afterwards be rubbed over with the very finest sandpaper, when entirely dry.

The finish of this work must not be a full glaze, but the “smear” glaze, described in the last chapter. A full glaze would fill up the lines, and so injure the appearance of the work. In order that the black lines may be effective, the ground of the decoration should be rather light in color. The best colors for the body of wares to be decorated in this way are light cream color, or a mixture of cream color and red clay, producing a light brick-red; or, if they can be procured, any light colored clays which have been artificially tinted in the body. A colored ground for this work can be procured by having the piece of ware dipped in colored slips. The design is then incised upon the surface, after it has become dry enough, in the same manner as before. The lines are cut through to the body of the ware beneath, showing a color contrasting with that upon the surface with good effect.

Work of this kind is also executed upon stoneware. That of Miss Hannah Barlow will be remembered as an instance of incised work, in black lines, upon a red or light brown body. Incised work also appears upon the Lambeth stoneware. It is here used upon grey stoneware, with raised dots in white, and colored with blue glaze. Incised decorations can also be executed upon earthenware in this way. White ware may be used, and the design drawn upon it in the “green” state. The piece may then be finished with colored glaze, which sinks into the depressions formed by the lines, and so gives them a darker color than the rest of the surface. Carved work can also be introduced with good effect as an accessory to incised work, and a few words as to this kind of decoration will now be in place.

CARVING IN CLAY.

This work can be made very beautiful and effective by those familiar with the use of carving tools. The tools used should be short-handled carving tools, and the ware should be in the “green” state and as for incised work, neither too wet nor too dry. Dry enough for clean cutting with the tools, and yet not so dry as to be brittle. When in just the right state, clay forms a fine substance for carving. As it dries very rapidly, means should be used to keep it sufficiently moist until the work is finished.

The outline can be drawn upon the clay with a sharp-pointed modeling tool, and the carving can be executed with gouges, chisels, and parting tools of the various shapes and sizes necessary. Veining can be done with a pointed modeling tool, and dots can be readily incised by pressure on the surface. Raised dots can be made with softer clay which has been dipped up in a quill of the required size, which is then pressed upon the surface at the place where the dot is desired.

Carved work can be finished by the use of sand paper, a camel’s hair brush, and water, or a sponge; but these should not be used very freely, as they will injure the sharpness and cleanness of the cutting, which it is desirable to preserve. Carving looks best when finished with a “smear” glaze.

Another variety of carved work can be produced upon a vase which has been previously dipped in colored “slip.” The design is then cut through the covering to the ground beneath, which should be of a contrasting color.

Another method by which many varied and beautiful effects can be produced is that of incising lines, and cutting or pressing designs into the soft clay, which are then filled with various colored slips. The lines or other designs should be deeply cut, moistened with water and filled with the colored clays, which should be pressed into the depressions and allowed to dry. When dry, the surface can be scraped in a similar manner to that made use of in finishing incised work. It should be noted that in this latter work the body of the clay should be in a softer state than would be necessary for other carved or incised work.

Carving in clay, as well as modeling and incising can be glazed with the colored majolica glazes with excellent effect.

CHAPTER VII.

PAINTING UPON THE BISCUIT.



AFTER the ware has been fired once and is in what is called the “biscuit,” it can be decorated by the use of underglaze colors, in the form of thin washes resembling water-color painting. Ware used for this purpose should be made of white, or very light cream-colored clay, as the colors, being used in transparent washes, would be dimmed, and would have their tints sullied by use upon a ground of dark color.

On white ware those colors should be used which will best stand the heat necessary in the firing and glazing of this ware. For this reason painting executed on white is usually in monochrome, and only the strongest colors are used, such as blue, black, or brown. The best effect is produced by the use of blue. This style of work is familiar from the numerous pieces of Japanese ware, of porcelain or white earthen-ware, painted in blue, which are brought to this country.

The method of painting is simple and not likely to prove difficult or uncertain as to its results in the hands of one familiar with the use of water colors. The color generally used is dark blue, and there are several different blues which furnish very fine tints. Among these might be mentioned Emery’s cobalt blue, which is excellent and does very well upon the white ware manufactured here. Phillips’ mazarine blue and Harrison’s Persian blue are also very rich, deep colors, the latter having a purplish tint. One of the best blues, as reliable and pleasing in color as any, has been prepared by Mr. Joseph Bailey, of the Hamilton Road Pottery.

The color must be rubbed down until perfectly smooth, and should be mixed with water and enough gum arabic to keep it from rubbing off in the necessary handling before firing. The design having been sketched upon the ware to be decorated, with a lead pencil, the first washes may be put in with a light tint of the blue. These should be laid on with a free hand and without retouching before they have dried. The darker tints may then be laid in, and lastly the deepest shadows. These should be painted thickly and sharply to

give the proper depth of color. If necessary they may be painted over a second time; care should be taken, however, not to have an excess of paint in any part, as that would prevent the glaze from adhering to the ware. The lines should be particularly clear and sharp, as they generally spread more or less in the glazing, and if they are not well defined and the shadows are not dark enough, the effect will be blurred and indistinct. In this kind of work clearness and distinctness are to be desired. If the painting is carefully executed and the various degrees of light and shade are well preserved, something of the effect of a painting in water colors will be produced. Monochromes in brown and black are done in the same way, only that in the case of the latter color it is better to mix a very little blue with it, as that will make it stand the fire better. Before glazing, the ware should be fired once to "harden on" the colors. If the color should run in the glazing, the painting may be repaired by the use of gilding over the glaze, by which the blurred outlines may be concealed. When, from the perfection of the firing, this is unnecessary, work of this kind, and especially that in blue, may be very much enhanced by a judicious use of gilding.

PAINTING IN THE LAMBETH OR BENNETT STYLE.

This is done upon a very light, cream-colored ware. The method of painting is similar to that described above, with the exception that all colors can be used. In using the weaker colors, and in mixtures, the same rules that have been given before must be observed. Colors which do not stand the fire very well, such as yellow and rose color, must be laid on thickly, while in mixtures, as well as when used alone, the stronger colors should be used sparingly. In the Lambeth or Bennett style of painting upon faience, the designs are either wholly or partly conventional. The grounds are either left the natural color of the ware or painted. The designs usually of flowers are in natural colors very slightly shaded and surrounded by a dark outline.

The design having been drawn upon the surface with a lead pencil, the ground, if it is to be colored, may be laid on with some color, or mixture of colors, which will harmonize with the decoration. As it is impossible to paint it smoothly with a brush, or to do it by any of the methods in use for over-glaze painting, the ground must be "dabbled" on, either with a small blending brush, or better, a sponge, which has been cut to the proper shape

and size, and which may be fastened in a quill and used as a brush. This is dipped in the color which has been mixed upon the palette, and with it the color is applied to the surface of the vase, producing a mottled ground, which may be shaded from light to dark, by the application of more or less color, as desired, and also may be varied by the introduction of other colors which will harmonize with each other, and with the decoration. The design may then be painted in flat washes, which should be laid on freely, with the shadows boldly defined. This done, a line, either of black or dark olive green, made from dark brown and a little green, may be painted around the design, outlining and relieving it against the ground.

The painting finished, the piece should then be fired to fix the colors. If necessary, it can be touched up in any places which are thought to need deeper tints after this firing, and before the final process of glazing. It must be remembered, however, that the effect of the glazing will be to deepen the colors, and, unless the result of this process has been learned from experience, the tendency will rather be to get them too dark than too light. The most important thing is, to keep the relative value of the lights and shadows. If this is done, the effect will not be bad, even if the colors should fade a little in firing.

Another very important item in under-glaze painting, which has been mentioned before, is, that the colors must be in proper condition for use, and to that end, must be carefully rubbed down with a muller or palette knife. Inattention to this detail has caused much work of rude and rough appearance to be produced, which might otherwise have been fairly good.

In the Lambeth and Bennett ware the charming effect of color will generally be found to have been produced by a system of burning which would scarcely be considered legitimate, according to the strict rules of the potter's art. The ware is not fired in the biscuit as hard as such ware is usually fired, and the glaze applied is softer than the body of the ware demands. This soft glaze is used to save the colors, and not being suited to the body of the ware will, generally, be found to be badly crazed. In this case, however, lovers of beautiful and harmonious colors, in the decoration of pottery, will think that the ends justifies the means, and as the articles are wholly adapted for ornamental purposes, this defect of glaze detracts but little from their value.

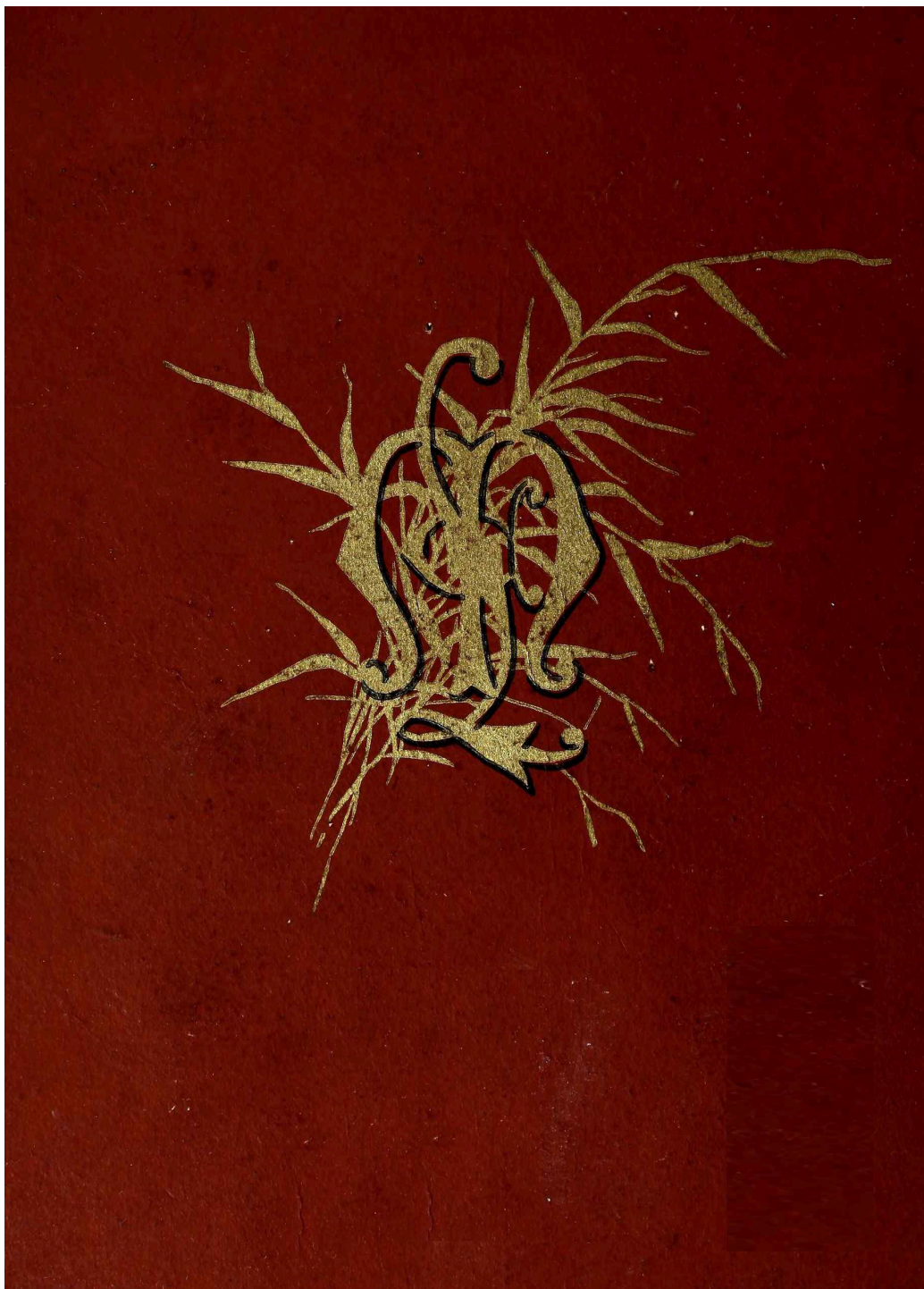
Some work of a similar character has recently been done in this city, however, in which the ware has been fired and glazed in such a manner as to make it perfectly durable, and, at the same time, great beauty of coloring has been retained. This result is not only important from the fact that it offers opportunity for the production of ornamental wares, but, also, from the consideration that the perfection of the firing and glazing of the ware renders it valuable for articles of use. Nothing could be prettier than sets of tableware made of this cream-tinted clay, decorated in colored, conventional designs. This would, we think, be the perfection of ware for ordinary use, and would form, with the combination of the delicately tinted body and the harmonious colors of the underglaze decoration, a service both pleasing to the eye and very durable.

It is not the province of a work on under-glaze painting to enter into the subject of over-glaze decoration, yet as they are frequently combined, mention may be made of some of the varieties of decoration in which the two processes are used. The most common form of the union of the two methods is that in which gilding is used as an adjunct to under-glaze decoration. Gold is always applied over the glaze, as it would be utterly destroyed by the action of the heat necessary in the firing of any hard glaze. That prepared by decorators by dissolving gold in aqua regia may be used, or the “bright” gold, which is obtained in a liquid state. The former is the most durable, and when fired can be burnished or left with a dead finish, which in many cases has the most pleasing effect. The “bright” gold, as its name indicates, becomes bright in the firing and the burnishing is thus rendered unnecessary. On a “smear” glaze the “bright” gold produces a beautiful dead gold effect, owing to the dead surface of the ware to which it is applied. In Japanese wares, dark blue, under-glaze, often appears in conjunction with gold, and also with red, green, or other colors which have been put on over the glaze.

One beautiful result of the combination of over and under-glaze painting is seen in the ware painted by Lemonnier. In this the ground is generally painted under the glaze in a manner similar to that described in this chapter for the production of Lambeth faience. The design is then painted with over-glaze “relief” colors, and the high lights laid in heavily, the whole producing an effect which combines the delicacy of the over-glaze colors with the depth of the under-glaze.

We have tried to indicate, in the preceding chapters, the principal methods employed in the decoration of pottery under the glaze. These various methods are combined with each other and with over-glaze work in so many ways that it is difficult for any but an expert to distinguish in a given piece of ware the many processes which may have combined to produce the result. This renders the subject more difficult to treat comprehensively, and these various ramifications of the art of under-glaze painting can here be but indicated. It is hoped, however, that the suggestions given may be sufficient to afford the student of ceramic decoration an idea of the various forms and of the possibilities of the art.





Transcriber's Notes

- pg 44 Changed: glaze sufficiently soft to perserve
to: glaze sufficiently soft to preserve
- pg 49 Changed: may be added a drop of salycilic acid
to: may be added a drop of salicylic acid
- pg 57 Changed: in colored slips, and the thicknesss
to: in colored slips, and the thickness
- pg 92 Changed: delicately tinted body and the harmonions
to: delicately tinted body and the harmonious
- Table of Contents was created by the transcriber.

*** END OF THE PROJECT GUTENBERG EBOOK POTTERY
DECORATION UNDER THE GLAZE ***

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